

MASK WHAT MATTERS: CONTROLLABLE TEXT-GUIDED MASKING FOR SELF-SUPERVISED MEDICAL IMAGE ANALYSIS

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ABSTRACT

Annotation scarcity in specialized domains such as medical imaging is a fundamental barrier to the training of robust vision models. Self-supervised masked image modeling (MIM) provides a principled route around this barrier; prevailing formulations, however, rest on random, high-ratio masking that is simultaneously computationally inefficient and semantically misaligned with the very regions that govern downstream analysis. Region-aware variants narrow this gap only partially: by grounding their masking in reconstruction heuristics or task-specific supervised signals, they sacrifice the adaptability required to generalize across tasks and imaging modalities. We introduce Mask What Matters (MWM), a controllable text-guided masking framework for self-supervised medical image analysis. MWM repurposes pretrained vision-language models to perform prompt-based, zero-shot localization of task-relevant anatomical structures, and applies spatially differentiated masking that concentrates the reconstruction signal on diagnostically salient regions while suppressing redundancy in background areas. This controllable design directly aligns the pretraining objective with the downstream objectives, sharpens the learned representations, and transfers across tasks and modalities without recourse to per-image reports or labels. A comprehensive evaluation across three complementary medical imaging modalities—brain MRI, chest CT, and lung X-ray—demonstrates that MWM consistently surpasses strong MIM baselines such as SparK, improving classification accuracy by up to +3.1 percentage points and, in detection and instance segmentation, raising box average precision (BoxAP) by +1.3 and mask average precision (MaskAP) by +1.1. Critically, these gains are attained at a substantially lower overall masking ratio (e.g., 40% versus 70%), establishing that semantic targeting, rather than aggressive masking, is the decisive driver of representation quality. Collectively, these findings establish controllable, text-driven masking as an effective and broadly generalizable principle for semantically aligned self-supervised learning, and chart a concrete path toward robust vision models for medical image analysis.

1 INTRODUCTION

Image classification, segmentation, and object detection constitute the foundational tasks of computer vision, and over the past decade deep learning has become the principal engine of progress across all three. This progress, however, is predicated on the availability of large-scale annotated datasets. In specialized domains such as medical image analysis, high-quality annotations are scarce, costly, and inseparable from expert labor; this annotation bottleneck remains one of the primary barriers to the widespread clinical deployment of deep learning.

Self-supervised learning (SSL) addresses this bottleneck directly by exploiting unlabeled data for pretraining, enabling models to acquire transferable feature representations in the absence of any manual annotation [Jaiswal et al. (2020); Liu et al. (2021); Krishnan et al. (2022); Huang et al. (2023)]. Among the SSL paradigms, masked image modeling (MIM) has become the dominant approach. By masking portions of an image and requiring the model to reconstruct the missing content, MIM compels the encoder to infer structural and semantic regularities from the visible context; the representations it yields transfer strongly across a broad range of downstream tasks,

with medical image analysis prominent among them [He et al. (2022); Gupta et al. (2025); Xiao et al. (2023)].

Prevailing MIM approaches—masked autoencoders (MAE) [He et al. (2022)] and SparK [Tian et al. (2023)] among them—typically adopt random masking strategies that are, by construction, unaligned with the regions of interest that determine downstream performance. This task-agnostic masking entails two principal limitations. First, the learned representations are prone to semantic misalignment with the downstream objectives, curtailing transfer performance. Second, to raise the probability of masking informative content, existing methods resort to extremely high masking ratios (e.g., 75% in MAE), inflating reconstruction difficulty and thereby demanding large-scale data and substantial computational resources. Natural language, by contrast, exhibits compact structure and high semantic density, and supports effective learning even at low masking ratios [Devlin et al. (2019)]. The reliance on high masking ratios in image-based MIM stems largely from the conjunction of inherent information redundancy and the absence of semantic-aware masking—a combination that prevents models from concentrating on the regions that genuinely matter.

These deficiencies are especially acute in medical imaging, where two properties compound them. First, medical images exhibit pronounced semantic sparsity: diagnostically relevant information is typically confined to small, localized regions—lesions or organs, for instance—whereas the majority of the image comprises semantically redundant background. Second, the field spans diverse modalities—MRI, CT, and X-ray—across which the shape, size, and spatial distribution of task-relevant areas vary substantially from task to task. Jointly, these properties impose stringent requirements on both the efficiency and the generalizability of any masking strategy.

To meet these requirements, we introduce a controllable, text-guided masking framework, hereafter **Mask-What-Matters (MWM)**. Given a user-specified, open-vocabulary description or phrase, a pretrained vision–language model supplies zero-shot localization cues, which we convert into robust regions of interest (ROIs). MWM then applies region-specific, prompt-conditioned masking—assigning higher ratios to semantically important areas (e.g., lesions and organs) and lower ratios to background—thereby injecting downstream semantics directly into the pretraining signal, and doing so without any per-image reports or labels.

Our contributions are threefold:

- **A controllable text-guided masking framework for medical imaging—the first of its kind.** MWM couples a vision–language model (BiomedCLIP) with segmentation-based refinement (SAM) to localize task-relevant regions from open-vocabulary prompts and to apply differentiated ROI-versus-background masking, thereby resolving the semantic misalignment intrinsic to random masking rather than merely mitigating it.
- **An annotation-free, backbone-agnostic design.** MWM dispenses with per-image reports and labels entirely, and integrates into both ViT- and ConvNet-based MIM pipelines under a single unified protocol, so that its semantic guidance is realized without additional supervision or backbone-specific engineering.
- **Consistent gains across modalities and tasks.** On brain MRI, chest CT, and X-ray datasets, MWM surpasses state-of-the-art MIM baselines such as SparK, improving classification, detection, and segmentation alike, and does so at markedly lower masking ratios—establishing text-driven generative pretraining as an effective and generalizable principle for medical imaging.

2 RELATED WORK

2.1 MASKED IMAGE MODELING

Masked modeling originated in natural language processing with the bidirectional encoder representations from transformers (BERT) model [Devlin et al. (2019)], which learns contextual representations by masking tokens and predicting them during pretraining. Adapting this principle, [He et al. (2022)] introduced MIM to computer vision, masking portions of an image during pretraining and tasking the model with reconstructing the missing regions. This paradigm generalizes strongly across both the natural and medical image domains [Gupta et al. (2025); Liu et al. (2021); Huang et al. (2023)]. Representative methods include MAE [He et al. (2022)], MedMAE [Gupta et al. (2025); Xiao et al. (2023)], and SparK [Tian et al. (2023)]: MAE employs high-ratio random masking to steer ViTs

toward high-level semantic representations; MedMAE transfers this idea to medical images through domain-specific adaptations; and SparK reformulates the strategy for convolutional backbones, exploiting their intrinsic multi-scale features to enhance pretraining effectiveness.

For all these advances, conventional MIM approaches remain wedded to random or structure-agnostic masking strategies that disregard the semantic requirements of downstream tasks. Such non-selective masking leads the model to overemphasize irrelevant regions during pretraining, eroding transfer effectiveness. To counter this, recent work has explored region-aware masking strategies intended to improve semantic alignment, proceeding along two principal directions.

The first line of work introduces loss-driven adaptive masking: methods such as hard patches mining (HPM) [Wang et al. (2023a)] and its extension AnatoMask [Li et al. (2024)] dynamically analyze reconstruction errors across regions and assign higher masking probabilities to areas of larger loss, seeking to prioritize semantically valuable structures. This criterion, however, is liable to misdirect the model toward hard-to-reconstruct yet task-irrelevant regions. In brain magnetic resonance imaging (MRI), for instance, structurally complex regions such as the spinal cord or the eyeballs may attract disproportionate attention despite being unrelated to downstream objectives such as tumor detection.

The second line of work incorporates external perceptual modules to guide masking. FocusMAE [Basu et al. (2024)], for example, leverages pretrained Region Proposal Networks (RPNs) to identify high-information regions for selective masking. Although effective on specific tasks, this strategy requires supervised training of RPNs on domain-specific annotated data, a dependency that constrains its cross-task generalization.

Collectively, these limitations expose a clear need for a masking mechanism that is at once more flexible, semantically aligned, and generalizable. Natural language, a rich and interpretable source of supervision, offers a compelling avenue for guiding masked modeling in a manner that is at once task-aware and annotation-free.

2.2 VISION-LANGUAGE PRE-TRAINED MODELS

The rapid maturation of vision-language alignment models has made it increasingly feasible to perceive and identify task-relevant regions accurately under zero-shot and open-world settings [Radford et al. (2021)]. Foundational approaches such as the Contrastive Language-Image Pretraining (CLIP) family induce a unified embedding space for images and text through large-scale joint pretraining on image-text pairs, and have been widely adopted for zero-shot and open-set visual tasks. Within medical imaging, extensions such as BiomedCLIP [Zhang et al. (2023)] further sharpen cross-modal understanding, delivering robust performance across diverse downstream applications. Yet, although vision-language pretrained models are now pervasive across supervised learning tasks, the prevailing body of work remains confined to downstream applications such as object detection and semantic segmentation [Zhong et al. (2022); Aleem et al. (2024); Koleilat et al. (2024)], leaving open the question of how their open-vocabulary and zero-shot capabilities might benefit masked image modeling.

In sum, recent region-aware MIM approaches—whether dynamic masking driven by reconstruction loss or masking guided by external perception modules—improve semantic focus only partially, and remain encumbered by recurring limitations: dependence on additional supervision, misalignment between masked regions and task semantics, and limited flexibility. In parallel, and despite the strong downstream performance of vision-language models, little effort has been directed at incorporating natural language prompts into the MIM pretraining phase to furnish task-aware masking guidance. These observations converge on a clear and promising direction, which the present work pursues: harnessing vision-language semantic guidance to enable efficient and generalizable masked image modeling.

3 METHODOLOGY

3.1 OVERALL FRAMEWORK

As illustrated in Figure 1, **Mask What Matters (MWM)** is organized into two core stages: (1) text-guided region localization and (2) region-aware masked image modeling. The first stage employs

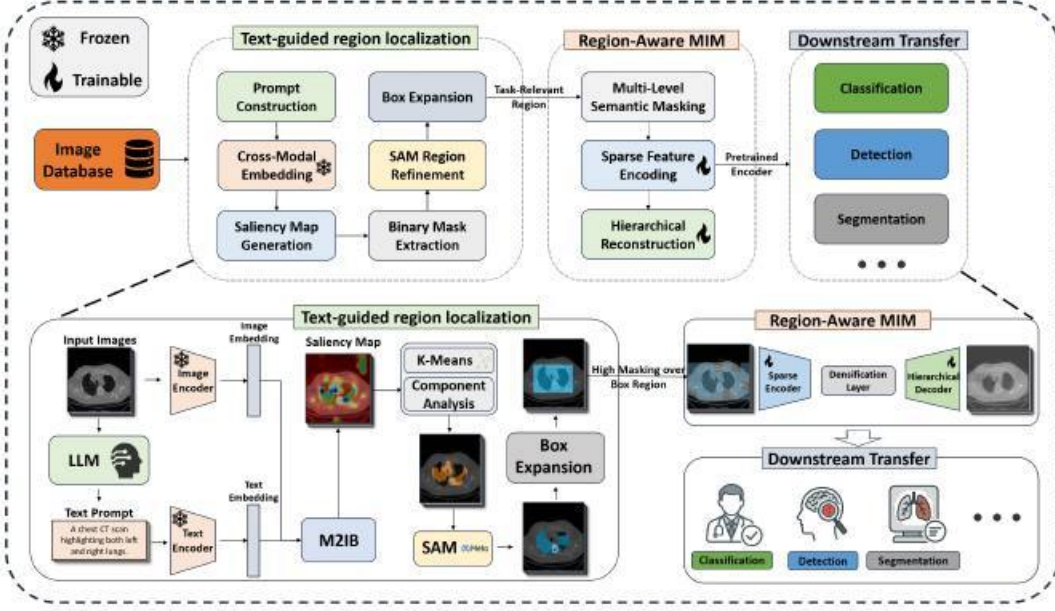


Figure 1: Overview of the MWM framework. The top panel depicts the three-stage pipeline: (1) text-guided region localization, (2) region-aware masked image modeling, and (3) downstream transfer. The bottom panel details how textual prompts drive region localization and the ensuing differentiated, region-conditioned masking.

a frozen vision–language model to identify task-relevant regions from an open-vocabulary prompt; the second stage applies differentiated masking ratios across these regions to steer the reconstruction process, thereby conditioning the spatial distribution of the reconstruction signal on the localized semantics.

3.2 TEXT-GUIDED REGION LOCALIZATION

MWM localizes task-relevant regions through a multi-stage pipeline whose stages compose into a single spatial guide. The pipeline begins with LLM-based prompt generation and employs BiomedCLIP to extract cross-modal embeddings; from these embeddings it derives a saliency map via the Multi-Modal Information Bottleneck (M2IB), binarizes the map with K-Means, and refines the result using the Segment Anything Model (SAM). Finally, the refined mask is converted into a bounding box with a controllable expansion margin, yielding a spatial guide that is robust to residual boundary noise for the subsequent masked modeling stage.

Prompt Generation. In medical image analysis, the design of the textual prompt is a decisive determinant of localization quality. To strengthen semantic guidance, we query large language models (LLMs) with a structured template of the form “Describe the typical visual characteristics of a [category] in a [modality] image.” Alongside these full-sentence prompts, we admit concise category phrases as an alternative input modality, and we report a detailed comparison of the two prompt styles in the experimental section.

Cross-Modal Embedding. Once the prompt is generated, a BiomedCLIP model [Zhang et al. \(2023\)](#) encodes the medical image and its prompt into a shared semantic space. Given an image I and a prompt T , the embeddings are computed as:

$$Z_{\text{img}} = \Phi_{\text{img}}(I), \quad Z_{\text{text}} = \Phi_{\text{text}}(T) \quad (1)$$

where Φ_{img} and Φ_{text} denote the vision and language encoders, respectively. These cross-modal embeddings constitute the semantic foundation on which saliency estimation is built in the subsequent stages.

Saliency Map Generation. A cross-modal saliency estimation module grounded in the Multi-modal Information Bottleneck (M2IB) [Wang et al. \(2023b\)](#) localizes task-relevant visual regions by align-

ing the image and text embeddings while suppressing modality-specific redundancy. Formally, the saliency map $\lambda_S \in [0, 1]^{H \times W}$ assigns to each spatial location an importance weight proportional to its semantic relevance to the prompt. M2IB obtains this map by optimizing the following objective:

$$\mathcal{L}_{\text{M2IB}} = \text{MI}(Z_{\text{img}}, Z_{\text{text}}) - \gamma \cdot \text{MI}(Z_{\text{img}}, I) \quad (2)$$

where Z_{img} and Z_{text} are the image and text embeddings encoded by BiomedCLIP, and γ governs the trade-off between preserving cross-modal relevance and filtering out task-irrelevant visual information.

Binary Mask Extraction. Given the saliency map, we binarize it via unsupervised K-Means clustering [Lloyd (1982)] to delineate the region indicated by the text prompt, yielding a preliminary foreground mask. Concretely, the map is partitioned into two pixel clusters, and the cluster with the higher saliency values is designated foreground. To refine this mask and suppress spurious activations, we then apply connected-component analysis to the binary map and retain only the largest foreground regions. This procedure filters out noisy edge areas and concentrates attention on the core semantic structures.

Region Refinement. To sharpen spatial accuracy, we incorporate SAM as a refinement module [Kirillov et al. (2023)]. For each selected connected region $c_i \in \mathcal{C}^*$, we compute its minimal enclosing bounding box and supply it as the visual prompt to SAM.

Given the original image I and the set of bounding boxes, SAM predicts a refined segmentation mask:

$$M_{\text{SAM}} = \text{SAM}(I, \text{Box}(c_i)) \quad (3)$$

The resulting mask M_{SAM} affords a more faithful spatial delineation of the target structure.

Box-Based Region Expansion. The SAM mask produced in the previous stage retains residual boundary noise that can misdirect the model’s focus during representation learning. To render the region guidance robust to such noise, we convert the refined SAM mask into a bounding box with a controllable expansion margin, which then serves as the final region of interest (ROI) for downstream masked modeling.

3.3 REGION-AWARE MASKED IMAGE MODELING

Given the region of interest produced by the localization module in Section 3.2, MWM performs masked image modeling through a three-stage pipeline: (1) a multi-level masking strategy that assigns differentiated masking ratios according to region importance, (2) a sparse encoder that processes only the visible (unmasked) patches, and (3) a hierarchical decoder that reconstructs the full image from the sparse multi-scale features.

Multi-Level Semantic Masking. MWM employs a multi-level masking strategy that operationalizes the semantic regions identified by the text-driven localization process. Patches falling within the regions highlighted by the textual prompts—those associated with tumors or organs—are masked at a high ratio, compelling the model to reconstruct, and therefore to represent, the task-relevant semantic content. Background regions, by contrast, are masked lightly or left visible, thereby curtailing redundant computation. Figure 2 contrasts the resulting masking patterns with those of conventional random masking.

Sparse Feature Encoding. The visible patches are aggregated into a sparse image and encoded following the SparK design [Tian et al. (2023)]:

$$\begin{aligned} X_{\text{sparse}} &= \{x_i \mid m_i = 1\}, \\ F &= \Phi_{\text{sparse}}(X_{\text{sparse}}) \end{aligned} \quad (4)$$

where x_i is the i -th patch, $m_i \in \{0, 1\}$ indicates visibility, Φ_{sparse} is the encoder, and $F = \{F_1, F_2, F_3, F_4\}$ are the resulting multi-scale features. Although we instantiate MWM with con-

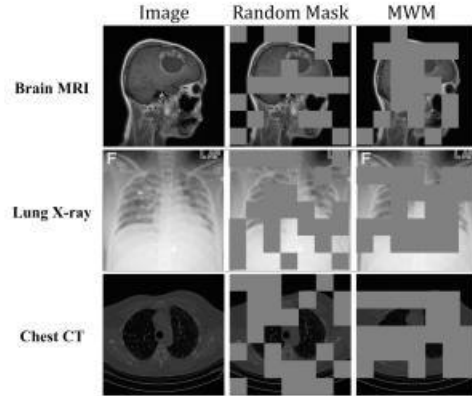


Figure 2: Comparison of masking strategies. Gray blocks denote masked regions.

volitional backbones, the same interface extends to ViTs through token indexing; it is precisely this modality-agnostic interface that renders the design backbone-agnostic.

Hierarchical Reconstruction. A lightweight UNet-style decoder reconstructs the full image in a top-down manner. At each scale l , the masked locations are populated with a learned embedding M_l :

$$F'_l(i) = \begin{cases} F_l(i), & \text{if patch } i \text{ is visible,} \\ M_l, & \text{if patch } i \text{ is masked.} \end{cases} \quad (5)$$

The densified map F'_l is projected by a scale-specific layer ϕ_l to obtain D_l , which is subsequently upsampled and fused with the features at the next lower level:

$$D_{l-1} = B_{l-1}(D_l) + \phi_{l-1}(F'_{l-1}). \quad (6)$$

Finally, the model is trained with mean-squared error on masked patches:

$$\mathcal{L}_{\text{recon}} = \frac{1}{|\mathcal{I}_{\text{mask}}|} \sum_{i \in \mathcal{I}_{\text{mask}}} \|\hat{x}_i - x_i\|^2, \quad (7)$$

where $\mathcal{I}_{\text{mask}}$ denotes the set of masked patch indices, x_i the original patch, and $\hat{x}_i$ its reconstruction; the loss is therefore evaluated exclusively over the masked locations.

4 EXPERIMENTS

4.1 DATASETS

Self-Supervised Pretraining Datasets. Self-supervised pretraining draws on three datasets that together span distinct imaging modalities. (1) *Brain MRI* comprises approximately 17,000 images drawn from Brain Tumor MRI [Cheng et al. (2015)], BRISC [Fateh et al. (2025)], and BraTS 2018 [Menze et al. (2014)], covering a range of tumor types and anatomical structures. (2) *Lung X-ray* consists of roughly 39,000 chest radiographs from COVID-QU-Ex [Rahman et al. (2021); Chowdhury et al. (2020)], spanning COVID-19, non-COVID pneumonia, and normal cases. (3) *Chest CT* comprises 12,000 slices from a lung-disease dataset [Konya (2020)], accompanied by segmentation masks of fibrotic lungs from 107 patients.

Downstream Fine-Tuning Datasets. Downstream evaluation, spanning classification and detection, employs three further datasets. (1) *Brain Tumor MRI* [Feltrin (2023)] contains 4,479 MRI images spanning 44 tumor types, including astrocytoma, glioblastoma, meningioma, and ependymoma. (2) *Pediatric Lung X-ray* [Kermanny et al. (2018)] consists of 5,863 pediatric chest X-rays labeled as either normal or pneumonia. (3) *Chest CT Cancer* [Hany (2020)] comprises CT images across four classes: adenocarcinoma, large cell carcinoma, squamous cell carcinoma, and normal tissue. Together, these datasets exercise MWM across heterogeneous modalities, anatomies, and label granularities, from binary screening to fine-grained 44-way tumor classification.

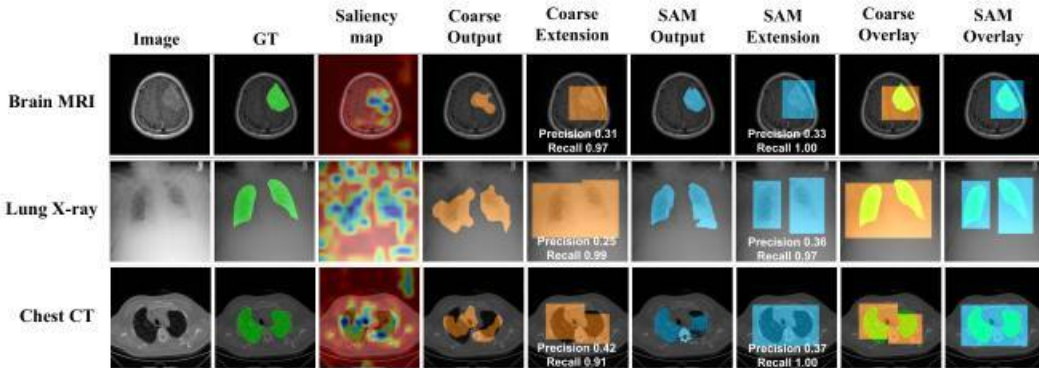


Figure 3: Representative results of text-guided region localization across all three imaging modalities, illustrating consistent identification of task-relevant structures despite pronounced modality-specific anatomical variation.

Method	Brain MRI		Chest CT		Lung X-ray	
	Prec.	Rec.	Prec.	Rec.	Prec.	Rec.
<i>Prompt Type</i>						
Phrase	0.22	0.86	0.21	0.99	0.42	0.82
Descrip. Sent.	0.30	0.94	0.37	0.88	0.44	0.66
<i>Impact of SAM</i>						
Coarse Ext.	0.22	0.95	0.36	0.87	0.37	0.84
+ SAM Ext.	0.30	0.94	0.37	0.88	0.42	0.82

Table 1: Text-guided localization performance (occlusion precision/recall) across prompt types and refinement strategies, reported for each of the three modalities. *Coarse Ext.* denotes bounding-box extension of the coarse region; + *SAM Ext.* denotes subsequent refinement with SAM. Within each block and modality, bold marks the stronger setting; across every modality, well-chosen prompts sustain high recall while SAM refinement improves precision.

4.2 EXPERIMENTAL SETUP

To ensure a controlled comparison, all self-supervised methods use backbones of comparable model scale: convolution-based methods adopt ResNet-50 (He et al., 2016), and transformer-based methods use ViT-S (Dosovitskiy et al., 2020). The input resolution is fixed at 224×224 , and all models are initialized from ImageNet-pretrained weights. Pretraining uses an initial learning rate of 2×10^{-4} , and downstream fine-tuning preserves both the encoder and the input resolution. All experiments are conducted on NVIDIA Tesla A100 cards with 80 GB VRAM under Python 3.9, Ubuntu 22.04.3, and PyTorch 1.10.0. Unless stated otherwise, non-target variables (e.g., prompt type or SAM usage) are held at their empirically optimal settings during evaluation, so that each reported comparison isolates the single factor under study.

4.3 EVALUATION OF TEXT-GUIDED LOCALIZATION

We first assess the text-guided localization module across all three imaging modalities—brain MRI, lung X-ray, and chest CT—and then isolate the respective influences of prompt type and SAM refinement. Predicted bounding boxes are compared against expert-annotated segmentation masks, and we report occlusion precision ($|R_p \cap R_{gt}|/|R_p|$) and occlusion recall ($|R_p \cap R_{gt}|/|R_{gt}|$), where R_p is the predicted region and R_{gt} the ground-truth annotation.

As Table 1 shows, text-guided localization with well-chosen prompts attains consistently high occlusion recall across every dataset (around 0.82) while sustaining competitive precision. These results establish that the framework identifies task-relevant regions from textual descriptions alone, and that it does so robustly across pronounced modality-specific anatomical variation, as the representative visualizations in Figure 3 confirm.

Effect of Prompt Type on Region Localization. To characterize the role of language design, we contrast the two prompt types. As reported in Table 1, descriptive sentence prompts are superior on Brain MRI and Chest CT, confirming that richer semantics aid localization. Concise phrases, by contrast, prevail on lung radiographs in recall (0.82 vs. 0.66), plausibly because the broad anatomical coverage of the lungs renders general terms already sufficient. Prompt granularity thus interacts systematically with anatomy, and the optimal choice tracks the spatial extent of the target structure—a dependence that MWM exploits rather than suffers, since both prompt styles are admissible inputs.

Effect of SAM Refinement on Region Localization. We further quantify the contribution of spatial refinement by applying SAM to the coarse masks. As Table 1 shows, SAM refinement raises precision while preserving high recall: on Brain MRI, precision improves from 0.22 to 0.30 with recall held above 0.94. SAM therefore tightens the localized regions without sacrificing coverage of the target.

4.4 COMPARISON WITH PREVIOUS SSL METHODS

We next evaluate the generalization and representation capacity of MWM against representative self-supervised learning (SSL) methods drawn from both the reconstruction-based and contrastive paradigms. To guarantee a fair comparison, every method is pretrained and evaluated under identical datasets and training settings, so that any performance difference is attributable to the masking strategy rather than to disparities in data or optimization.

Image Classification. We examine classification performance on the three downstream medical imaging datasets under two standard protocols: (1) *full fine-tuning*, in which the entire model is updated, and (2) *linear probing*, in which the encoder is frozen and only a linear classifier is trained on top of it.

- **Full fine-tuning:** Table 2 reports classification accuracy for MWM alongside several representative self-supervised methods—MAE He et al. (2022), SparK Tian et al. (2023), AnatoMask Li et al. (2024), SimCLR Chen et al. (2020a), MoCoV2 Chen et al. (2020b), and BYOL Grill et al. (2020). MWM attains the highest accuracy on every modality, evidencing robust generalization across heterogeneous imaging conditions. The margin is largest on Chest CT, where MWM reaches 97.5% accuracy—exceeding the MIM baselines SparK by +3.1 and AnatoMask by +1.7 points. This gain follows directly from text-guided masking, which concentrates the reconstruction signal on semantically critical yet spatially sparse structures, such as the lung lobes in CT, while withholding it from redundant background.
- **Linear probing:** To isolate the intrinsic quality of the learned representations, we conduct linear-probing experiments in which the encoder is frozen, comparing MWM against the two strongest recent masked image modeling methods, SparK and AnatoMask. As Table 3 shows, MWM outperforms both on all three datasets by clear margins—for example, +3.5 points over SparK on Chest CT and +6.9 on Lung X-ray—confirming that text-driven pretraining yields more discriminative features even when the encoder is held fixed, and thus that the advantage originates in representation quality rather than in downstream adaptation.

Object Detection and Instance Segmentation. To determine whether the learned representations generalize beyond classification, we evaluate brain-tumor detection and instance segmentation on the BR35H dataset Hamada (2025). As Table 4 shows, MWM attains the highest AP^{box} (46.9%) and the second-highest AP^{mask} (45.9%), surpassing SparK by +1.3 and +1.1 points respectively. MWM thus transfers effectively to both object-level localization and pixel-wise segmentation, carrying its advantage from image-level recognition through to dense prediction.

Taken together, these results establish that incorporating text-driven masking during pretraining equips the model with more informative and more transferable representations. Because this semantic guidance improves performance across classification, detection, and segmentation alike, it constitutes a general principle for visual representation learning rather than a task-specific expedient.

Method	Type	Brain MRI	Chest CT	Lung X-ray
MoCoV2	CL	95.3	91.7	93.1
BYOL	CL	91.3	93.7	93.9
SimCLR	CL	94.5	93.1	94.1
MAE	MIM	95.4	91.7	94.9
AnatoMask	MIM	96.5	95.8	95.2
SparK	MIM	96.2	94.4	94.7
MWM(Ours)	MIM	96.8	97.5	96.0

Table 2: Full fine-tuning classification accuracy (%) on the three downstream datasets, comparing MWM against contrastive (CL) and masked-image-modeling (MIM) baselines under identical pre-training and evaluation settings. The best result in each column is shown in bold; MWM ranks first on every modality, with its largest margin on Chest CT.

Method	Brain MRI	Chest CT	Lung X-ray
AnatoMask	70.2	63.9	85.3
SparK	69.8	63.5	81.6
MWM(Ours)	71.3	67.0	88.5

Table 3: Linear-probing accuracy (%) on the downstream datasets with a frozen encoder, evaluated against the two strongest MIM baselines. Best per column in bold; MWM leads on all three modalities by clear margins, indicating that its advantage resides in the learned representation itself rather than in fine-tuning.

Method	Detection		Segmentation	
	AP^{box}	AP_{75}^{box}	AP^{mask}	AP_{75}^{mask}
AnatoMask	45.2	53.1	46.2	54.1
SparK	45.6	53.2	44.8	52.0
MWM (Ours)	46.9	53.5	45.9	54.1

Table 4: Object detection (AP^{box} , AP_{75}^{box}) and instance segmentation (AP^{mask} , AP_{75}^{mask}) results (%) on the BR35H dataset, comparing MWM against the two strongest MIM baselines. Best per column in bold; MWM leads on both detection metrics and on AP_{75}^{mask} , extending its advantage from image-level recognition to dense prediction.

4.5 ABLATION STUDY

We further conduct ablation experiments that isolate the individual contributions of prompt design, SAM-based region refinement, and masking ratio to MWM’s downstream classification performance.

4.5.1 EFFECT OF PROMPT DESIGN AND REGION REFINEMENT

As Table 5 shows, descriptive sentence prompts outperform short phrases on brain MRI and chest CT, whereas the ordering reverses on lung X-ray. This pattern mirrors the localization results of Section 4.3: prompts that yield more accurate region identification also yield stronger downstream accuracy, establishing a direct link between localization fidelity and representation quality. Decisively, both prompt types—irrespective of granularity—outperform random masking on every dataset. This establishes that the gains stem from text-guided semantic alignment between the pretraining and downstream objectives, not from the particulars of any single prompt formulation. The SAM ablation reinforces this account: incorporating the refinement module during pretraining consistently improves downstream classification across all three modalities, corroborating its role in sharpening the text-guided masking signal.

4.5.2 EFFECT OF MASKING RATIO

As Figure 4 shows, MWM attains 96.8% classification accuracy at a masking ratio of merely 40%—well below the roughly 70% conventionally adopted in masked image modeling—and, despite this reduced ratio, exceeds SparK’s best configuration at 70% masking by +2.4 points in absolute accuracy. To exclude the possibility that SparK’s weaker result reflects a suboptimal masking configuration rather than a genuine gap, we evaluate SparK at 40%, 50%, 60%, and 70%; the 70% setting indeed yields SparK’s highest accuracy, confirming that the comparison

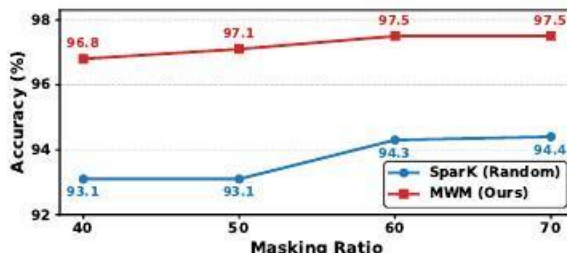


Figure 4: Classification accuracy on the Chest CT dataset as a function of masking strategy and masking ratio. MWM peaks at a 40% ratio, surpassing the best SparK configuration (70%) while masking far less of the image.

Group	Setting	Brain MRI	Chest CT	Lung X-ray
Prompt Type	No Prompt	96.2	94.4	94.7
	Phrase	96.4	96.5	96.0
	Sentence	96.8	97.5	95.2
Impact of SAM	w/o SAM	96.4	96.8	95.8
	w/ SAM	96.8	97.5	96.0

Table 5: Downstream classification accuracy (%) under varying prompt types and region-refinement strategies across the three imaging modalities. *No Prompt* denotes random masking without text-guided localization and thus serves as the baseline within each block. *w/* and *w/o* denote with and without SAM, respectively. Best per column in bold. Both prompt types surpass the no-prompt baseline on every modality, and adding SAM improves accuracy throughout.

is made against SparK’s own optimum. These findings overturn the prevailing assumption that high masking ratios are inherently optimal for masked image modeling. They establish instead the decisive advantage of semantically guided, task-aware masking: by preserving task-relevant content while discarding redundancy, MWM delivers stronger downstream performance at substantially lower masking ratios.

5 CONCLUSION AND LIMITATIONS

This work presents **Mask What Matters (MWM)**, a self-supervised pretraining framework that unifies text-guided semantic localization with region-aware masking. By using natural language prompts to highlight task-relevant areas and applying differentiated masking across them, MWM realizes semantically aligned representation learning without any per-image annotation.

Comprehensive experiments establish the effectiveness and generalizability of MWM: (1) natural language prompts reliably guide semantic region localization, validating text-driven masking as a sound design principle; (2) MWM consistently outperforms strong existing methods such as SparK and AnatoMask in classification across all three imaging modalities, and delivers further gains in detection and instance segmentation; and (3) MWM sustains, indeed improves, its performance at markedly lower masking ratios (e.g., 40%), demonstrating that semantic guidance, rather than aggressive masking, is what drives representation quality. Together these findings establish controllable, text-driven masking as a general and generalizable principle for semantically aligned self-supervised learning.

Limitations. For all its demonstrated strengths, the framework leaves two questions open. First, although text prompts afford flexible guidance, we have not yet systematically characterized MWM’s robustness to variation in prompt style, vocabulary, or noise. Second, the reliance on external textual descriptions introduces a dependency that may attenuate generalization where reliable prompts are unavailable or inconsistent. These are bounded, well-scoped extensions of the present study rather than obstacles to its central conclusions.

Looking ahead, text-driven, region-aware masking charts a principled path toward more semantically grounded self-supervised learning, and extending MWM beyond medical imaging offers a natural avenue for broadening the reach of vision–language pretraining.

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A USE OF LARGE LANGUAGE MODELS (LLMs)

We used LLMs *only* for language editing and presentation polishing (e.g., grammar, phrasing, and minor stylistic clarity) of text written by the authors. LLMs were *not* used for idea generation, experimental design, data annotation, code implementation, or analysis/interpretation of results. All technical content, algorithms, proofs, figures, tables, metrics, and conclusions were authored and verified by the authors. No confidential or identifying data were provided to LLMs beyond anonymized manuscript text.